

A price tracker that tells you what changed

Catalogue Watch checks a shop's shelf every morning, keeps every run, and writes a one-page catalog that leads with the biggest price drop.

The problem

A daily export of every product is a file nobody opens. A buyer needs to hear when a price drops or a book sells out, and nothing on the mornings when nothing moved.

What I built

- Crawler with two engines, fast HTTP and headless Chromium, that return identical rows
- Every run stored in SQLite, with a CSV export and 30 days of price history per product
- A catalog page where the biggest drop gets the biggest tile, a red strike and a sparkline
- One short alert email, sent only when something changed
- Retries, then a failed run recorded and reported without overwriting the last good prices

Measured results

1.0 s

to crawl 2 pages and 32 books on the live practice store

30 days

of history, including one failed run and one quiet morning

13 of 13

tests pass: parsing, each change type, the report, the alert and failed runs



The morning catalog page, run 30



The alert email

Recent runs					
Every run is kept, so a price can be traced back day by day.					
Date	Books	Changes	Time	Status	
Wed 30 Sep, 7:02 am	32	5	0.2 s	Done	
Tue 29 Sep, 7:02 am	32	0	0.3 s	Done	
Mon 28 Sep, 7:02 am	32	1	0.3 s	Done	
Sun 27 Sep, 7:02 am	32	0	0.3 s	Done	
Sat 26 Sep, 7:02 am	32	0	0.3 s	Done	
Fri 25 Sep, 7:02 am	32	1	0.3 s	Done	
Thu 24 Sep, 7:02 am	32	2	0.2 s	Done	
Wed 23 Sep, 7:02 am	32	0	0.3 s	Done	
Tue 22 Sep, 7:02 am	32	1	0.3 s	Done	
Mon 21 Sep, 7:02 am	32	1	0.2 s	Done	
Sun 20 Sep, 7:02 am	32	1	0.3 s	Done	
Sat 19 Sep, 7:02 am	32	1	0.3 s	Done	
Fri 18 Sep, 7:02 am	32	0	0.3 s	Done	
Thu 17 Sep, 7:02 am			0.2 s	Failed. The site did not answer (con	

Run history with the failed day

books.toscrape.com is a store made for scraping practice and never changes its prices. Day 1 crawls the live site; days 2 to 30 crawl local copies with price moves simulated, through the same code.

Stack

Python 3

httpx

BeautifulSoup

Playwright

SQLite

Jinja2

Code github.com/vanthienha199/catalog-watcher-demo